A VARIANT OF THE LOCAL CRITERION FOR FLATNESS VIA TOR ONE AND THE QUOTIENT BY AN IDEAL

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Vanishing of Tor one against the residue field implies flatness. Invoked in the first proof inside the span (lines 23976-23981). Reduces the claim to a Tor computation against the residue field.
- Tor one against a quotient as a kernel. Used in the second proof inside the span (lines 23990-24014). Reinterprets the hypothesis as injectivity of a multiplication map.

Source reference. Original lemma 00HK: sources/stacks-algebra/pinned-original-source.tex, source lines 9212–9212. Navigation pointer for \ref{lemma-flat-eq} used inside the counted span (line 23990); the argument that consumes it is retained in the span.

Source reference. Original lemma 00MK: sources/stacks-algebra/pinned-original-source.tex, source lines 23800–23800. Navigation pointer for \ref{lemma-local-criterion-flatness} used inside the counted span (line 23981); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 051C: sources/stacks-algebra/pinned-original-source.tex, source lines 23868–23868. Navigation pointer for \ref{lemma-what-does-it-mean} used inside the counted span (line 23977); the argument that consumes it is retained in the span.

Source reference. Original lemma 00M6: sources/stacks-algebra/pinned-original-source.tex, source lines 18907–18907. Navigation pointer for \ref{remark-Tor-ring-mod-ideal} used inside the counted span (line 24014); the argument that consumes it is retained in the span.

Lemma 0.1 (Variant of the local criterion)**.** *Let $R \to S$ be a local homomorphism of Noetherian local rings. Let $I \neq R$ be an ideal in R. Let M be a finite S-module. If $Tor_1^R(M, R/I) = 0$ and M/IM is flat over R/I, then M is flat over R.*

Proof. First proof: By Lemma 051C we see that $\mathrm{Tor}_1^R(\kappa, M)$ is zero where κ is the residue field of R. Hence we see that M is flat over R by Lemma 00MK.

Second proof: Let $\mathfrak{m}$ be the maximal ideal of R. We will show that $\mathfrak{m} \otimes_R M \to M$ is injective, and then apply Lemma 00MK. Suppose that $\sum f_i \otimes x_i \in \mathfrak{m} \otimes_R M$ and that $\sum f_i x_i = 0$ in M. By the equational criterion for flatness Lemma 00HK applied to M/IM over R/I we see there exist $\overline{a}_{ij} \in R/I$ and $\overline{y}_j \in M/IM$ such that $x_i \bmod IM = \sum_j \overline{a}_{ij}\overline{y}_j$ and $0 = \sum_i (f_i \bmod I)\overline{a}_{ij}$. Let $a_{ij} \in R$ be a lift of $\overline{a}_{ij}$ and similarly let $y_j \in M$ be a lift of $\overline{y}_j$. Then we see that

$$\begin{aligned} \sum f_i \otimes x_i &= \sum f_i \otimes x_i + \sum f_i a_{ij} \otimes y_j - \sum f_i \otimes a_{ij} y_j \\ &= \sum f_i \otimes (x_i - \sum a_{ij} y_j) + \sum (\sum f_i a_{ij}) \otimes y_j \end{aligned}$$

Since $x_i - \sum a_{ij} y_j \in IM$ and $\sum f_i a_{ij} \in I$ we see that there exists an element in $I \otimes_R M$ which maps to our given element $\sum f_i \otimes x_i$ in $\mathfrak{m} \otimes_R M$. But $I \otimes_R M \to M$ is injective by assumption (see Remark 00M6) and we win. $\square$